

## SECRET WEAPONS

**IN THE SUMMER OF 1944** London was subjected to a second blitz, this time by unpiloted V1 flying bombs. The Allies had been aware for some time that the Germans were developing "secret weapons," and devastating bombing raids both on the experimental center in Peenemunde, Germany, and on launch sites under construction in France delayed the V1's deployment. But by June 1944 the Germans had switched to smaller launchers that were not detected by Allied reconnaissance aircraft. Shortly after D-Day, Hitler ordered a flying-bomb offensive against London.

In summer 1944 about 100 V1s a day crossed the Channel, by day and night, and in all weather conditions. Although they were unarmed and unescorted, the flying bombs posed a novel challenge to air defenses.



### SPOILS OF WAR

*After the war's end, an American soldier studies a V2 in the underground rocket-assembly plant in Nordhausen, Germany. Slave laborers worked under appalling conditions to produce the rockets, and many thousands died of ill-treatment.*

### V1 AUTOPILOT

*This autopilot from a V1 flying bomb fed signals to the bomb's elevators and rudder to control altitude and direction. The terminal dive was initiated when a preset distance had been flown.*



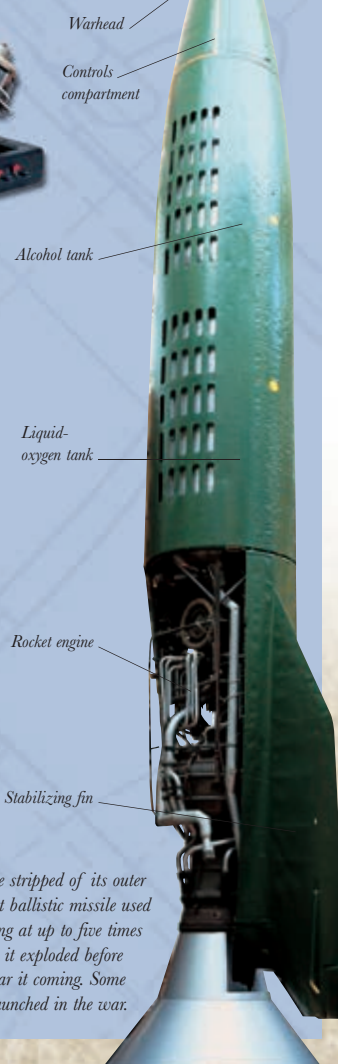
A combination of early-warning radar, AA guns, and fighters was deployed, as against any intruders, but Spitfires, Typhoons, and Mustangs found it hard to cope with such a fast, small target. Only the new high-performance Hawker Tempests could easily catch the flying bombs; jet-powered Gloster Meteors were also sent up as interceptors, the first operational use of jets by the Allies.

Shooting down a V1 was a hazardous action, since if it exploded it could easily destroy its attacker. Tempest pilots developed a technique of flying alongside a V1, lifting their wing under the flying bomb's wing, and tipping it over so that it spiraled down out of control.

After V1 launch sites in northern France were overrun by Allied troops in August 1944, the Germans began air-launching them from Heinkel He 111s flying in at low level over the North Sea. About half of the 8,000 V1s launched against Britain were shot down by aircraft or ground fire.

Hitler's other secret weapon, the V2 ballistic missile, was not interceptible, although Allied air forces made heroic efforts to destroy heavily defended V2 launch and production sites.

Fortunately, the Germans did not have an atomic warhead to put on the end of it. In all, the V weapons killed almost 9,000 people in Britain in 1944–45.



Warhead  
Controls compartment

Alcohol tank

Liquid-oxygen tank

Rocket engine

Stabilizing fin

### V2 ROCKET

*The V2 rocket, here stripped of its outer casing, was the first ballistic missile used in warfare. Traveling at up to five times the speed of sound, it exploded before its victims could hear it coming. Some 3,200 V2s were launched in the war.*

### FLYING-BOMB LAUNCH

*A German V1 flying bomb captured by the Allies at the end of the war – and bearing US insignia – is test-fired. Propelled by a primitive jet engine and flying at over 400mph (640kph), the V1 was packed with a ton of explosives. When it hit the ground, the destructive effect was impressive. Yet V1s killed only 5,475 people in Britain – less than one for each V1 launched.*